WORKED SOLUTION

STEP I 2016, Question 10

Mechanics • Particle collisions

STEP I 2016, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s1-q10

The key idea

There are three collisions, and each one is handled by the same two equations: conservation of momentum and Newton's law of restitution. The first two collisions happen at the same time but independently, so we can work them out separately to find the velocities of B and C just before they meet. The condition that B ends up at rest then gives one equation linking λ and e.

Solution

We model the four bodies as particles on a smooth table, so no horizontal forces act except in the collisions, and every collision is direct (along the line). Take the direction from A towards D as positive, so A starts with velocity u and D with velocity $-u$.



Part (i)

A **hits** B**.** Let A and B have velocities v_A and v_B afterwards. Then

$$\begin{aligned} \text{momentum:} \quad & \lambda m u = \lambda m v_A + m v_B, \\ \text{restitution:} \quad & v_B - v_A = eu. \end{aligned}$$

Substituting $v_B = v_A + eu$ into the first equation gives $(\lambda + 1)v_A = (\lambda - e)u$, so

$$v_A = \frac{(\lambda - e)u}{\lambda + 1}, \qquad v_B = v_A + eu = \frac{\lambda(1 + e)u}{\lambda + 1}.$$

D **hits** C**.** Let C and D have velocities v_C and v_D afterwards. Then

$$\begin{aligned} \text{momentum:} \quad & -mu = mv_C + mv_D, \\ \text{restitution:} \quad & v_D - v_C = e(0-(-u)) = eu. \end{aligned}$$

(Before the impact C and D approach at speed u; afterwards D is on the right of C, so they separate at rate $v_D - v_C$.) Solving these two equations:

$$v_C = -\tfrac{1}{2}(1+e)u, \qquad v_D = -\tfrac{1}{2}(1-e)u.$$

So C moves towards B with speed $\frac{1}{2}(1+e)u$, and D follows it to the left with the smaller speed $\frac{1}{2}(1-e)u$.

B **hits** C**.** Now B moves right at v_B and C moves left, so they approach with speed $v_B - v_C$. After the collision B is at rest; let C have velocity w. Then

$$\begin{aligned} \text{momentum:} \quad & mv_B + mv_C = 0 + mw, \\ \text{restitution:} \quad & w - 0 = e(v_B - v_C). \end{aligned}$$

Eliminating w gives $v_B + v_C = e(v_B - v_C)$, that is,

$$(1-e)v_B = -(1+e)v_C.$$

Substituting the velocities found above,

$$(1-e)\cdot\frac{\lambda(1+e)u}{\lambda+1} = (1+e)\cdot\tfrac{1}{2}(1+e)u.$$

We may divide by $(1+e)u$, since $u \neq 0$ and $1+e > 0$:

$$\begin{aligned} 2\lambda(1-e) &= (1+e)(\lambda+1) \\ 2\lambda - 2\lambda e &= \lambda + 1 + e\lambda + e \\ \lambda - 1 &= e(3\lambda+1), \end{aligned}$$

and as $3\lambda + 1 > 0$,

$$e = \frac{\lambda-1}{3\lambda+1},$$

as required.

For the deduction,

$$\frac{1}{3} - e = \frac{(3\lambda+1) - 3(\lambda-1)}{3(3\lambda+1)} = \frac{4}{3(3\lambda+1)} > 0,$$

since $\lambda > 1$. Hence $e < \frac{1}{3}$. (Also $e > 0$ because $\lambda > 1$, so the collisions are genuinely not perfectly inelastic.)

Part (ii)

After the third collision C has velocity

$$w = v_B + v_C = (1+e)u\left(\frac{\lambda}{\lambda+1} - \frac{1}{2}\right) = \frac{(1+e)(\lambda-1)u}{2(\lambda+1)},$$

which is positive because $\lambda > 1$: C now moves to the right, back towards D, which is still moving left with speed $\frac{1}{2}(1-e)u$.

Ask yourself Which equation does "the same speed" give, and how do we get rid of e?

The speeds are equal when

$$\frac{(1+e)(\lambda-1)}{2(\lambda+1)} = \frac{1-e}{2}.$$

From part (i), $1+e = \dfrac{4\lambda}{3\lambda+1}$ and $1-e = \dfrac{2(\lambda+1)}{3\lambda+1}$. Substituting and cancelling the common factor $\dfrac{1}{3\lambda+1}$:

$$\begin{aligned}\frac{4\lambda(\lambda-1)}{\lambda+1} &= 2(\lambda+1)\\ 2\lambda^2-2\lambda &= \lambda^2+2\lambda+1\\ \lambda^2-4\lambda-1 &= 0,\end{aligned}$$

so $\lambda = 2\pm\sqrt{5}$. Only $\lambda = 2+\sqrt{5}$ satisfies $\lambda > 1$ (the other root is negative). Then

$$e = \frac{1+\sqrt{5}}{7+3\sqrt{5}} = \frac{(1+\sqrt{5})(7-3\sqrt{5})}{49-45} = \frac{4\sqrt{5}-8}{4} = \sqrt{5}-2.$$

So

$$\lambda = 2+\sqrt{5}, \qquad e = \sqrt{5}-2.$$

Checks

- $e = \sqrt{5}-2 \approx 0.236$, which is indeed between 0 and $\frac{1}{3}$, as part (i) demands.
- With $\lambda = 2+\sqrt{5} \approx 4.236$, $u = 1$ and $e \approx 0.236$, running the three collisions numerically gives $v_B = 1$ (exactly, as it happens) and $v_C \approx -0.618$ before the third collision; afterwards B is at rest and C moves right at about 0.382, while D moves left at about 0.382. The common speed is $\frac{1}{2}(1-e)u = \frac{1}{2}(3-\sqrt{5})u$.
- Limiting case: as $\lambda \to 1$, $e \to 0$. With equal masses and $e = 0$, B and C meet with equal and opposite velocities, so coalescing leaves them at rest, as it should.

Common slips

- Getting the sign wrong in the restitution equation for the collision between C and D: write every velocity with the same positive direction and use "separation $= e\times$ approach" carefully.
- Forgetting that C arrives at the third collision moving *towards* B, so the approach speed is $v_B - v_C = v_B + \frac{1}{2}(1+e)u$, not $v_B - \frac{1}{2}(1+e)u$.
- Keeping the root $\lambda = 2-\sqrt{5}$, which is negative.

Beyond STEP

A collision with restitution is the simplest model of energy loss: the kinetic energy lost in a direct collision is proportional to $1 - e^2$, and $e = 1$ is the elastic case in which energy is conserved. In a first-year course these ideas lead to the centre-of-mass frame, where any two-body collision looks symmetric and the algebra above becomes almost trivial. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops collisions this way.

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